```
from google.colab import files
uploaded = files.upload()
```

Choose files Titanic-Dataset.csv

Titanic-Dataset.csv(text/csv) - 61194 bytes, last modified: 24/12/2021 - 100% done

```
Saving Titanic-Dataset.csv to Titanic-Dataset.csv
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# make plots look good
sns.set(style="whitegrid")

# LOAD DATASET
df = pd.read_csv("Titanic-Dataset.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	T
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	ST(3
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3

Next steps: Generate code with df New interactive sheet

```
# Basic info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

- Total rows
- Data types
- Missing values

```
# Statistical summary
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

- Mean age
- Max fare
- Distribution

```
# Target variable check
df['Survived'].value_counts()
```

Survived	count
0	549
1	342

dtype: int64

- How many survived vs died

```
# MISSING VALUES
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

- Age has missing values
- Cabin has many missing values
- Embarked has few missing values

```
# DATA CLEANING
```

```
df['Age'].fillna(df['Age'].mean(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

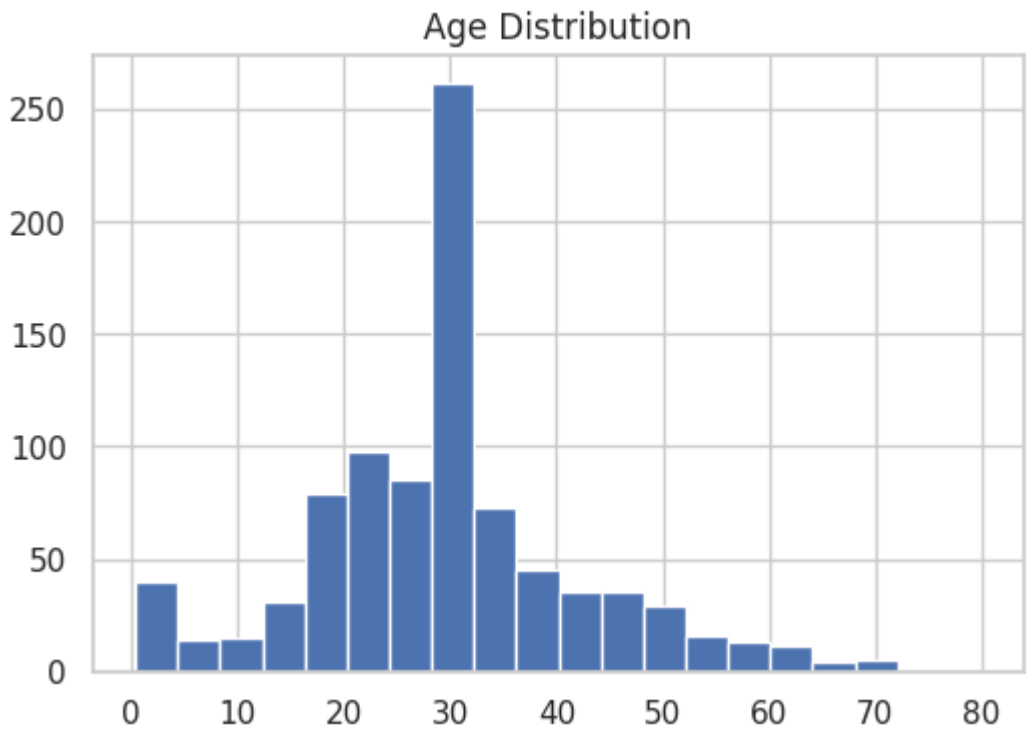
```
/tmp/ipykernel_3031/2861725518.py:1: FutureWarning: A value is trying t
The behavior will change in pandas 3.0. This inplace method will never

For example, when doing 'df[col].method(value, inplace=True)', try usi


  df['Age'].fillna(df['Age'].mean(), inplace=True)
```

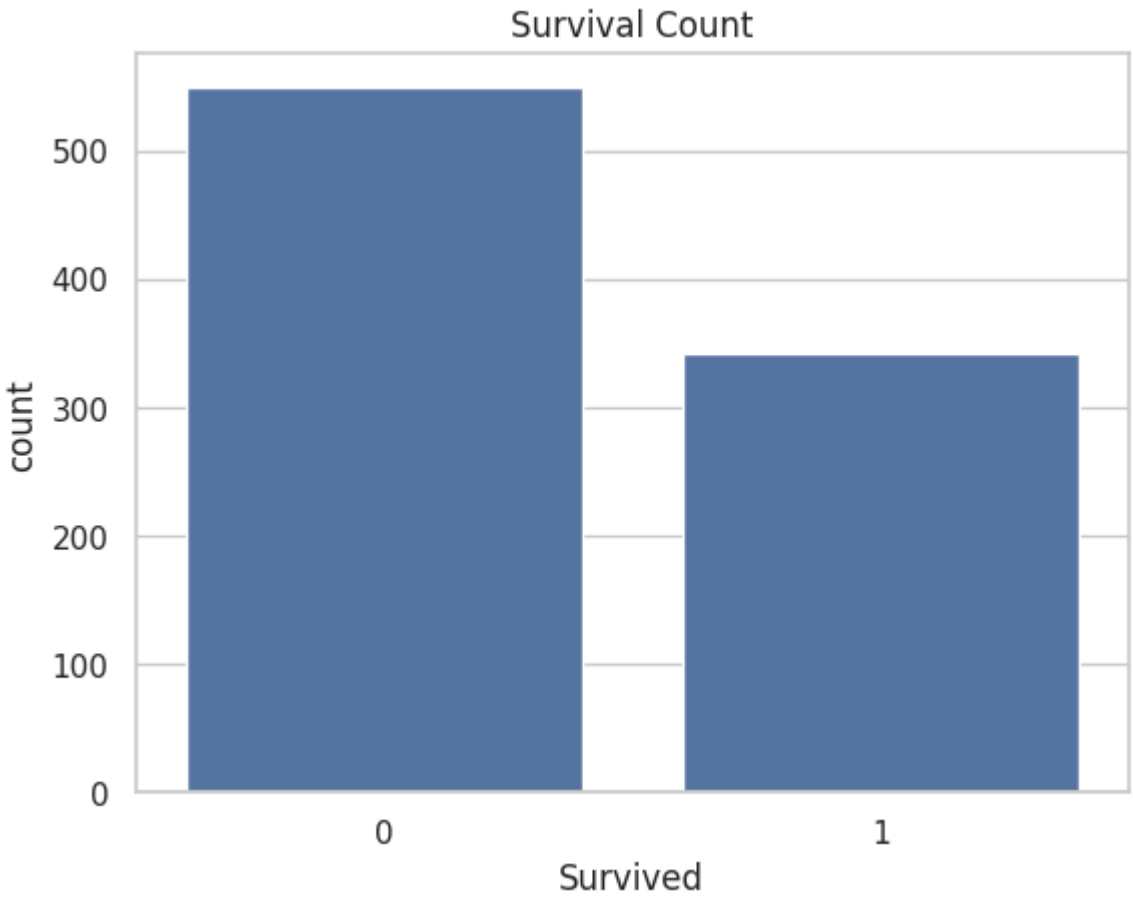
- Cabin can be ignored (too many missing)

```
# age distribution
plt.figure(figsize=(6,4))
df['Age'].hist(bins=20)
plt.title("Age Distribution")
plt.show()
```



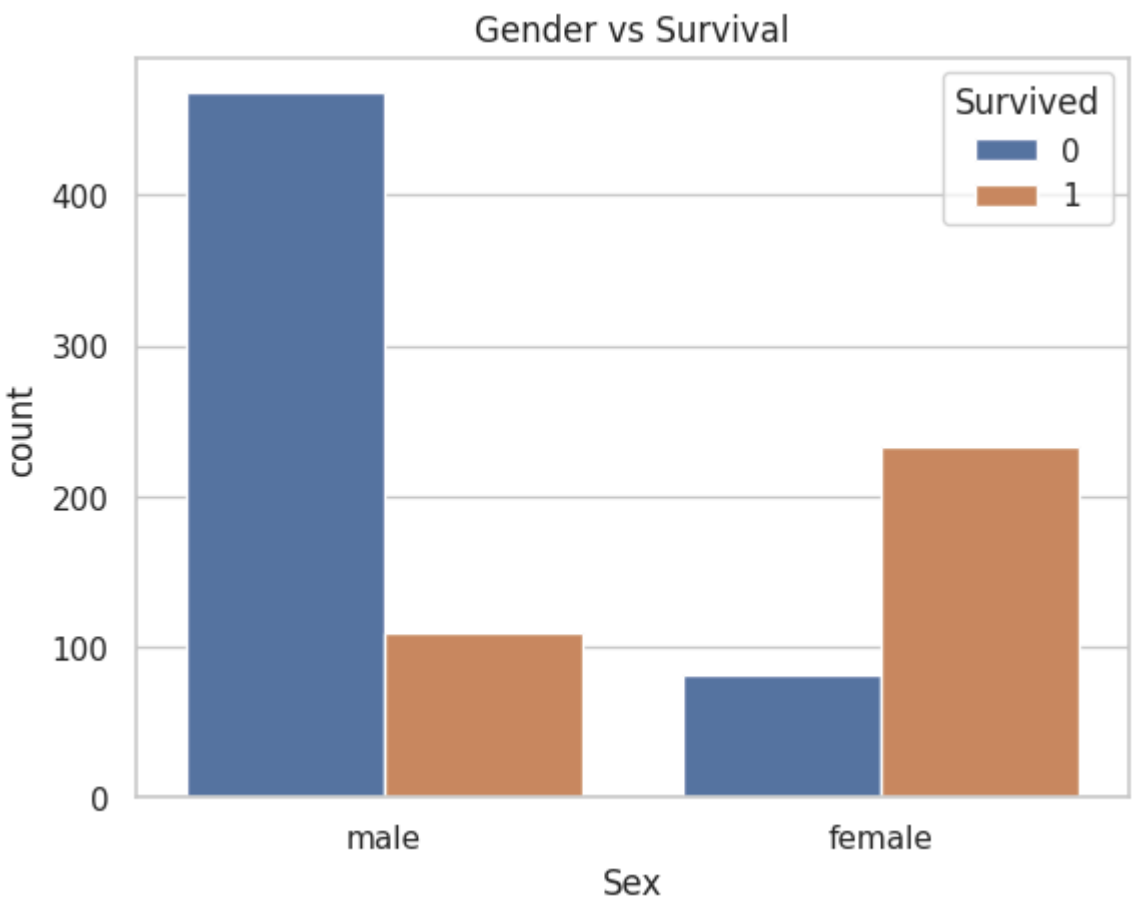
- Most passengers are between 20–40

```
# Survival count
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



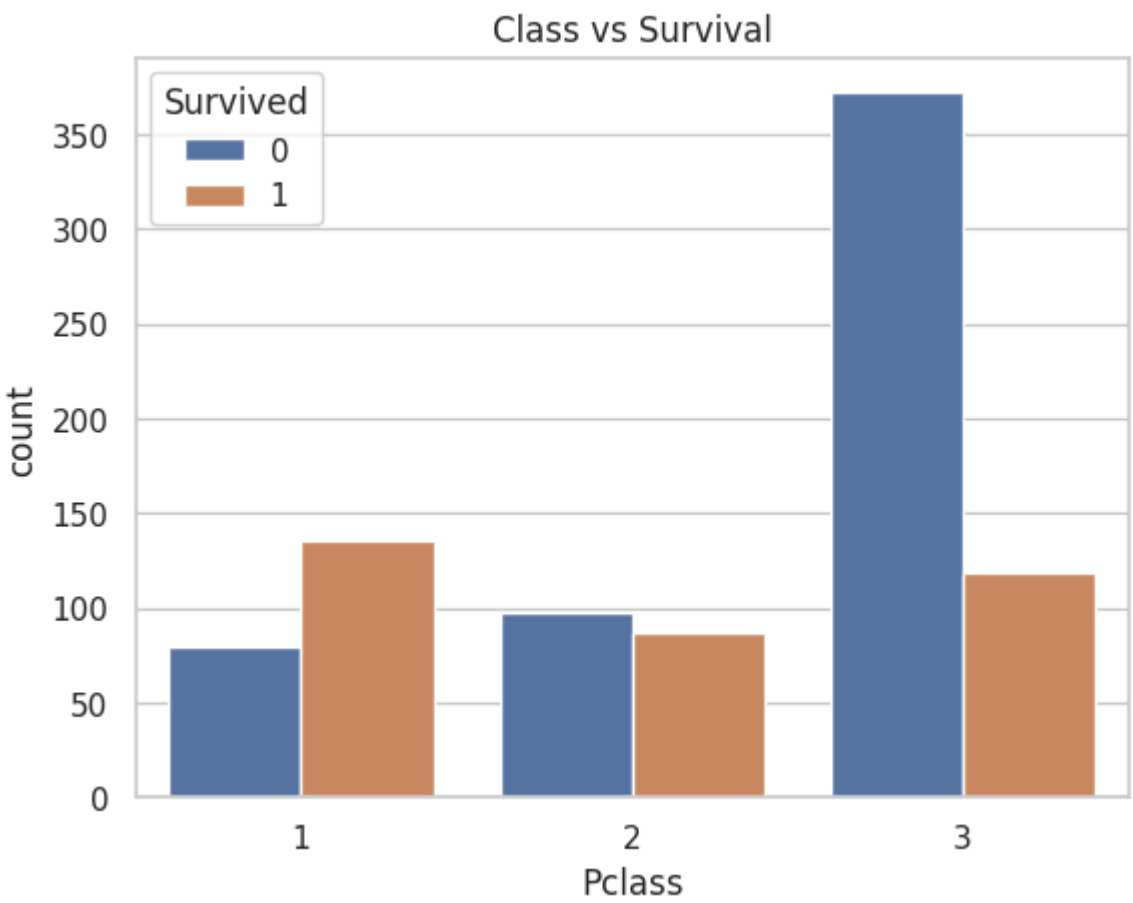
- Females survived more than males

```
# Gender vs Survival
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Gender vs Survival")
plt.show()
```



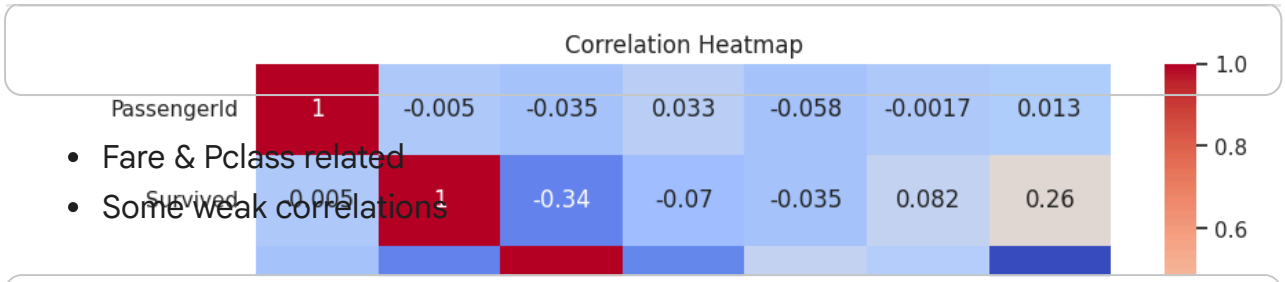
- Females survived more than males

```
# Class vs Survival
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Class vs Survival")
plt.show()
```

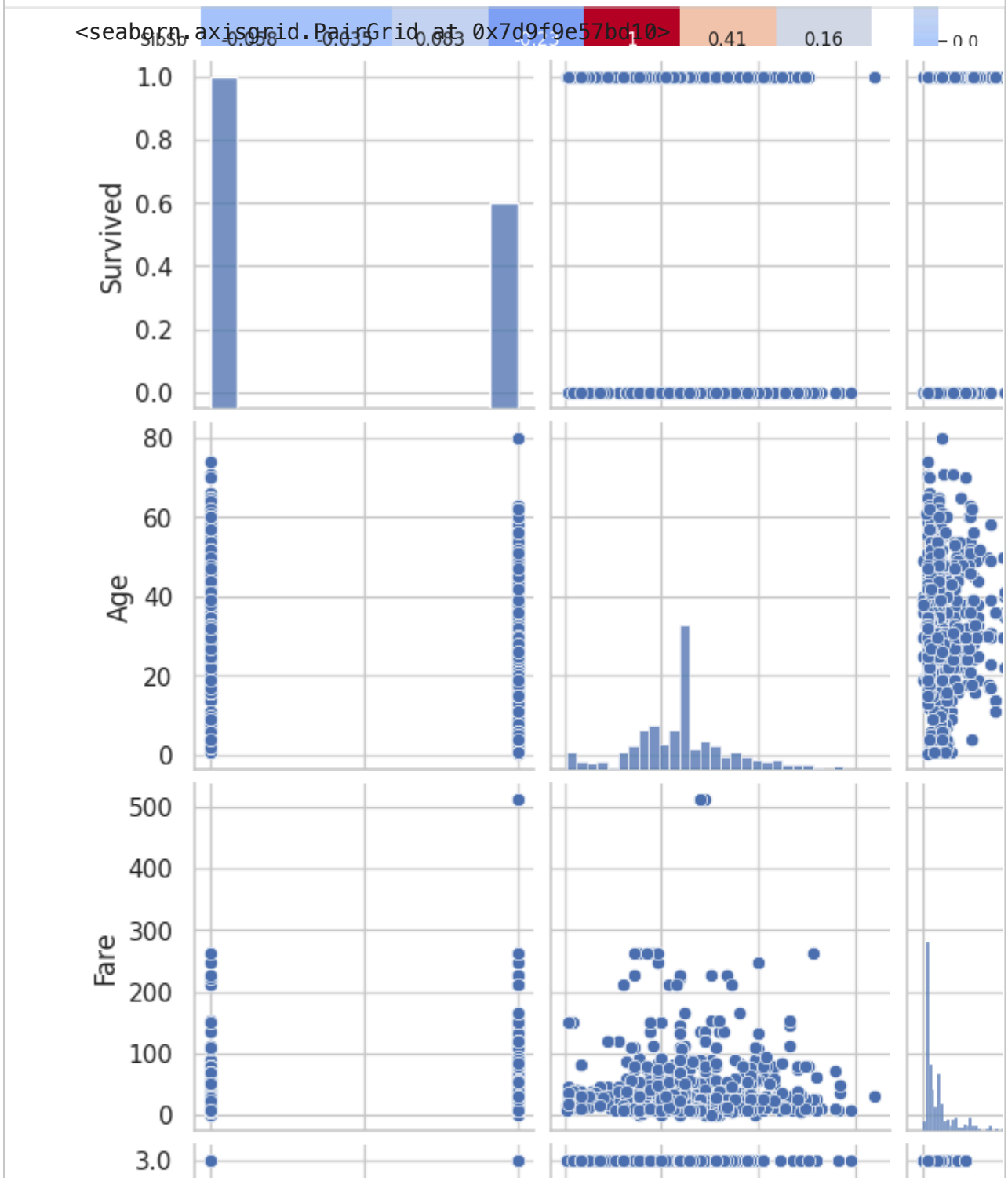


- 1st class survived more

```
# CORRELATION
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



- Fare & Pclass related
- Some weak correlations

```
#PAIRPLOT (ADVANCED)
sns.pairplot(df[['Survived','Age','Fare','Pclass']])
```

```
<seaborn.axisgrid.PairGrid at 0x7d9f9e57bd10>
```